National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 6 - Chapter 6	Page(s):	2

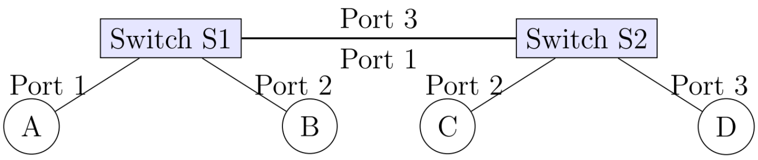
Student Name **Roll No.** **Section:**

Q1. Carefully read the Question before attempting: **[5 marks]**

Consider the switched LAN topology shown below. The switch tables are initially empty. The following sequence of frame transmissions occurs:

- Node A sends a frame to Node D.
- Node D replies with a frame to Node A.
- Node C sends a frame to Node D.

Show the state of the Switching Table for **Switch S2** after these three events. Format: (MAC Address, Interface). Also, show the same for **Switch S1**.



Solution:

- **Switch S1 Table:**

MAC Address	Interface
A	Port 1
D	Port 3

Note: S1 learns A from step 1. S1 learns D from step 2. S1 never sees the frame from C (step 3) because S2 knows D is on Port 3 and does not flood the frame back to S1.

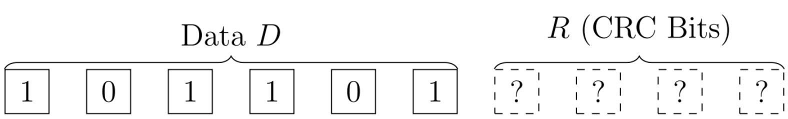
- **Switch S2 Table:**

MAC Address	Interface
A	Port 1
D	Port 3
C	Port 2

Note: S2 learns A from step 1 (incoming on Port 1). S2 learns D from step 2 (incoming on Port 3). S2 learns C from step 3 (incoming on Port 2).

Q2. Carefully read the Question before attempting: **[10 marks]**

Consider a data transmission scenario using a CRC generator polynomial $G(x) = x^4 + x + 1$. The data string to be transmitted is $D = 101101$.



1. Analytically calculate the 4-bit CRC remainder *R*. Show the long division in binary modulo-2 arithmetic. What is the actual bit-string transmitted by the sender?
2. A "burst error" of length k is a contiguous sequence of bits in which the first and last bits are errors, and the intermediate bits may or may not be errors. Can this specific generator *G(x)* can detect **all** burst errors of length *k = 3*. Give a reason?
3. Suppose a burst error occurs during transmission such that the received bit string has the 3rd and 4th bits (counting from the left, 1-based index) inverted. Does the receiver accept or reject this frame?

Solution:

1. **CRC Calculation:**
   - Generator $G(x) = x^4 + x + 1 \Rightarrow$ **10011** (5 bits).
   - Data $D = 101101$. Append 4 zeros: 101101**0000**.
   - Perform Binary Division (XOR):

```
                101101
              ----------
       10011 ) 1011010000
               10011
               -----
                01011
                00000
                -----
                 10110
                 10011
                 -----
                  01010
                  00000
                  -----
                   10100
                   10011
                   -----
                    01110
                    00000
                    -----
                    1110   <-- Remainder (R)
```

   The remainder R is **1110**.
   The transmitted string is 101101**1110**.

2. **Burst Error Detection:**
   Yes. A generator polynomial of degree r (here $r = 4$) detects **all** burst errors of length $k \leq r$. Since the burst length $k = 3$ and $3 \leq 4$, this error is guaranteed to be detected.

3. **Receiver Check:**
   The 3rd and 4th bits are inverted.
   Original Transmitted: 10**11**011110
   Received Frame: 10**00**011110
   Performing the division of 1000011110 by 10011:
   $1000011110 \div 10011$ yields a remainder of **1011**.
   Since the remainder is non-zero ($\neq 0000$), the receiver **rejects** the frame.